Zhaoxiang Cai

Preferred name: Simon

PhD (Cancer Data Science) · MBusA (First Class Honours) · BCompSc (First Class Honours)

CINSW Early Career Fellow · Senior Data Scientist (Cancer Data Science), Children's Medical Research Institute
Adjunct Lecturer, Faculty of Medicine and Health, University of Sydney

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1,273	9	8	5 of 10
TOTAL CITATIONS	H-INDEX	I10-INDEX	FIRST/CO-FIRST AUTHOR

Citation metrics from Google Scholar.

PROFESSIONAL APPOINTMENTS

Feb 2023 - present	**Senior Data Scientist (Cancer Data Science)**, Children's Medical Research Institute, Westmead
Jun 2026 - present	**Adjunct Lecturer**, Faculty of Medicine and Health, University of Sydney
Jun 2023 - Jun 2026	**Conjoint Associate Lecturer**, Faculty of Medicine and Health, University of Sydney
Mar 2020 - Feb 2023	**PhD Candidate (Cancer Data Science)**, Children's Medical Research Institute / University of Sydney
Jan 2019 - Mar 2020	**Data Scientist / Bioinformatician**, Children's Medical Research Institute, Westmead
Nov 2014 - Jan 2018	**Analyst Programmer**, Goldman Sachs, Melbourne

QUALIFICATIONS

Mar 2020 - Feb 2023 — **Doctor of Philosophy (Cancer Data Science)**, University of Sydney / Children's Medical Research Institute
Thesis: Large-Scale and Pan-Cancer Proteogenomic Analyses with Machine Learning

Jan 2018 - Dec 2018 — **Master of Business Analytics (First Class Honours)**, Melbourne Business School, University of Melbourne
- KPMG-MBS Data Challenge: 1st Prize Winner (Natural Language Processing)

Jan 2011 - Jul 2014 — **Bachelor of Computer Science (First Class Honours)**, Monash University
- Dux of Bachelor of Computer Science (highest overall ranking in graduating cohort)

GRANTS AND FELLOWSHIPS

Fellowships

2026 - 2028 — **Cancer Institute NSW Early Career Fellowship**
$597,732 · Sole Chief Investigator · Awarded 2025
Improving Cancer Diagnosis and Prognosis via Generative Proteomic Foundation Model and Federated Learning

Research Grants

2026 — **ResetData AI-F1 Competition**, national GPU compute award
$1,000,000 (in-kind GPU compute) · Application lead
ProCan Multimodal Cancer Foundation Model

2024 - 2026 — **Medical Research Future Fund (MRFF) Grant**
$993,500 · Associate Investigator
Targeting the Dysregulated Epigenome to Enhance Immunotherapy Response

Total competitive research funding contributed to: over $2.59 million (including $1 million in-kind GPU compute)

AWARDS AND HONOURS

2026	Cancer Research Network EMCR Outstanding Publication Award (Cai et al., Cancer Discovery, 2025)
2026	Cancer Bioinformatics Australia Symposium Travel Scholarship
2022	Sydney Cancer Partners PhD Scholarship
2020	CMRI Peter Rowe PhD Scholarship
2019, 2020	University of Sydney Innovation Challenge: 1st Prize ($7,500 each year)
2019	3rd Prize, ODIR-2019 International Computer Vision Competition (approx. $20,000)

SELECTED PUBLICATIONS

First and co-first author articles (5 of 10 peer-reviewed); full list at zhaoxiangsimoncai.github.io/research. Author name in bold; * denotes equal contribution. IF = journal impact factor; FWCI = Field-Weighted Citation Impact (Scopus/SciVal), above 1.0 indicates above world average.

1. **Cai, Z.**, Boys, E. L., Noor, Z., … Reddel, R. R. (2025). Federated deep learning enables cancer subtyping by proteomics. *Cancer Discovery*, OF1-OF16.
   First author · IF 29.5 · FWCI 3.34 · 15 citations
2. **Cai, Z.**, Apolinário, S., Baião, A. R., Pacini, C., Sousa, M. D., Vinga, S., … Gonçalves, E. (2024). Synthetic augmentation of cancer cell line multi-omic datasets using unsupervised deep learning. *Nature Communications*, 15(1), 1-12.
   First author · IF 18.1 · FWCI 3.38 · 47 citations
3. **Cai, Z.**, Poulos, R. C., Aref, A., Robinson, P. J., Reddel, R. R., & Zhong, Q. (2024). DeePathNet: a transformer-based deep learning model integrating multi-omic data with cancer pathways. *Cancer Research Communications*, 4(12), 3151-3164.
   First author · IF 4.0 · FWCI 4.08 · 43 citations
4. Gonçalves, E.*, Poulos, R. C.*, **Cai, Z.***, … Robinson, P., Zhong, Q., Garnett, M., & Reddel, R. R. (2022). Pan-cancer proteomic map of 949 human cell lines. *Cancer Cell*, 40(8), 835-849.
   Co-first author · IF 56.1 · FWCI 10.62 · 264 citations
5. **Cai, Z.**, Poulos, R. C., Liu, J., & Zhong, Q. (2022). Machine learning for multi-omics data integration in cancer. *iScience*, 103798.
   First author · IF 4.5 · FWCI 6.04 · 344 citations

SUPERVISION AND EXAMINATION

Jul 2025 - present	**Ms Fatemeh Mehdikhani**, PhD candidate Co-supervisor · Faculty of Medicine and Health, University of Sydney *Non-Invasive Diagnosis and Classification of Nevi and Early-Stage Melanoma*
Oct 2025	**Ms Rita Brito Gama**, Master of Philosophy External thesis examiner · University of Lisbon, Portugal *GAIN-DANN: A Domain-Adversarial Generative Model for Missing Data Imputation in Proteomics*

SELECTED PRESENTATIONS

2026	**Cancer Bioinformatics Australia Symposium**, Brisbane *Augmenting the Childhood Cancer Cell Map through Multi-Omic Data Synthesis and Predictive Modelling*
2024	**Advancing Multi-Omics into the Clinic Symposium**
2022	**70th ASMS Conference on Mass Spectrometry and Allied Topics**
2021	**KCA Precision Medicine for Childhood Cancer Symposium** **Invited**

SERVICE AND LEADERSHIP

2023 - present	**Leadership of Cancer Data Science Journal Club**, CMRI
2023 - present	**Manuscript peer review** Approx. five per year · Nature Communications, Genome Biology, Briefings in Bioinformatics
2025	**Invited external grant reviewer** Innovation and Technology Commission (Hong Kong) · Dutch Research Council (NWO, Netherlands)